

Markscheme

Mathematics: analysis and approaches

Practice question bank

8. (a)

Consider the expression $\left(\frac{8}{27}\right)^{-\frac{2}{3}}$.

Evaluate it exactly.

take the reciprocal for the negative power: $\left(\frac{8}{27}\right)^{-\frac{2}{3}} = \left(\frac{27}{8}\right)^{\frac{2}{3}}$ (M1)

take the cube root first: $\sqrt[3]{\frac{27}{8}} = \frac{3}{2}$ A1

square the result (M1)

$$\left(\frac{3}{2}\right)^2 = \frac{9}{4} = 2.25 \quad \text{A1}$$

[4 marks]

(b)

Hence find the value of $\left(\frac{8}{27}\right)^{-\frac{4}{3}}$.

note $-\frac{4}{3} = 2 \times \left(-\frac{2}{3}\right)$, so this is the square of the part (a) answer (M1)

$$\left(\frac{9}{4}\right)^2 = \frac{81}{16} = 5.0625 \quad \text{A1}$$

[2 marks]